**AMNS/BAH/720-SCB/0322-03/263315**

**DESIGN, ENGINEERING, SUPPLY, ERECTION, INSTALLATION, AUTOMATION & COMMISSIONING OF SAFETY FANCING FOR VARIOUS AM/NS LOCATIONS.**

**ArcelorMittal Nippon Steel India**

| | | | |
|---|---|---|---|
| **SAFETY STEEL MESH FENCING OF PROCESS LINE** | **AM/NS INDIA** | A joint venture between ArcelorMittal and Nippon Steel Corporation<br>ArcelorMittal NIPPON STEEL | **REFRENCE NO-** |

**SCOPE OF WORK FOR DESIGN, ENGINEERING, SUPPLY, ERECTION, INSTALLATION, AUTOMATION & COMMISSIONING OF STEEL MESH SAFETY FENCING & PIPE BARRICATION GAURDING OF PROCESS LINES, TRANSFER TROLLY & SOLAR PANEL AREA STEEL MESH SAFETY FENCING FOR ALL SSC**

**AS PER SOW ARCELORMITTAL NIPPON STEEL INDIA LTD BAHADURGARH.**

The following is the BOQ. In respect of terms & conditions, not explicitly provided in this Special Terms & Conditions, provisions of General Conditions of contract are applicable.

**Bidders are required to go through the tender document carefully before giving final quote for items mentioned in the attached list. If required, bidders can also visit our site for understanding the job properly; however, they should inform their program for site visit in advance to the contact persons (preferable technical contact 1).**

1. Complete Modular, Extra Heavy-Duty Machine Safety Fence for Machine Guarding.
2. The Perimeter Protection system consisting, uprights 60X60X2 mm and panel with frame 40x40x1.5 mm & fence grid of 22x100x3 mm.
3. Complied of safety standard ISO-13857: Safety distances to prevent the contact of dangerous Areas with upper and lower limbs
4. Complied of safety standard ISO 13854- EN 349: Minimum gaps to avoid crushing parts of the human body.
5. Impact test proven machine guarding fence as per ISO 14120.
6. Material: IS 2062 E250 & SAE 1008 Steel used to ensure the structure strength.
7. Pre-galvanized and powder coated safety fence.
8. Square Pole shade is standard of RAL: Yellow 1021, Steel Mesh Shade is standard of RAL: black 9005.
9. Zinc Plated Hardware with rubber capping of uprights.
10. Wedge type anchor fastener with Chemical Anchoring for the Uprights Fixing.
11. Anti-Climbed design of Wire Mesh.
12. CAD design to be submit for approval before manufacturing of product.
13. No cutting and Welding (Hot Work) Required.
14. Providing all labour for erection, installation, termination, commissioning for line wise safety fencing.
15. Providing additional accessories for Steel mesh safety fencing to complete project successfully.
16. Providing all required cables, wires, electric fitting, & fitting arrangement.

17. Vendor shall include service & commissioning team to & fro charges, conveyance charges, lodging boarding charges.
18. All work to be carried out as per safety norms which shall be guided to AM/NS team during work & no tolerance in safety will be accepted and penalty to be imposed if any violation.
19. Arranging the proper housekeeping after daily work completion & not leave the work site in an unsafe condition or any other condition that might cause injury to personnel, damage to existing work, plants or equipment.
20. Verify all installed equipment's before declared the work completion and should be certified by user.
21. Providing the individual equipment bill warranty period from date of installation.
22. Free servicing and repairing of installed Equipment's during warranty periods.

| Location | Job Description | Qty | Remark |
|---|---|---|---|
| Bahadurgarh SC | CRNCTL Line, CRS Line, CRWCTL Line, HRNCTL Line, HRS Line, Transfer trolley, Solar Inverter room fencing | 8 Set | • 1.5 Meter Height Steel mesh fencing with detachable MS Square section poles of 1.7 Meter Height & heavy foot plate for fastening, doors on both Drive and Non-Drive side, including Safety switch interlock, Lamp indications, Boom Barrier, Hooters and Light curtains interlocks.<br>• Electromagnetic actuators with interlocks for safety access control at doors.<br>• All fasteners should be "captive type" to avoid loss of fasteners during dismantling.<br>• 2" MS Pipe detachable barricading of 1.5 Meter Height with Hinge door at Uncoiler, Turnstile, Pit area equipment as frequent coil/packet handling |
| Pune SC | CRNCTL Line, CRS Line, CRWCTL Line, HRNCTL Line, HRS Line, HRWCTL Line, CR Mini CTL Line, CR Trapezoidal Line, Transfer trolley, Solar Inverter room fencing | 10 Set | |
| Indore SC | HRS Line, HRWCTL Line, Transfer trolley | 3 Set | |
| Chennai SC | CRNCTL Line, CRS Line, CRWCTL Line, HRNCTL Line, HRWCTL Line, HRS Line, Transfer trolley, Solar Inverter room fencing | 9 Set | |

| | | | |
|---|---|---|---|
| | | | will be done through crane.<br>• 3” MS Pipe detachable barricading of 1 Meter Height for coil transfer trolley with 2 Nos 1000 mm Boom Barrier for entrance as frequent coil handling will be done through crane.<br>• Impact capacity of 2000 N with 100 KG load travelling from 20 Kmph from 90 degree angle for fencing & 4000 N force for pipe barricading<br>• Safety Fencing of 1.5 Mtr Height with 2 Nos hinge doors for Solar inverter room |

**BOQ FORMAT FOR DESIGN, ENGINEERING, SUPPLY, ERECTION, INSTALLATION, COMMISSIOING OF STEEL MESH SAFETY FENCING FOR PROCESS LINES OF BAHADURGARH SC**

| SR. NO. | LINE | PRODUCT | QTY | UNIT |
|---|---|---|---|---|
| **Cold Rolled Slitter Line** | | | | |
| 1 | CRS | Steel Mesh Safety Fence | 76 | Mtr |
| 2 | CRS | 2" MS Pipe barricaded structure | 32 | Mtr |
| 3 | CRS | Safety hinge door | 6 | Set |
| 4 | CRS | Safety Magnetic switch | 6 | Set |
| 5 | CRS | Boom barrier 3000 mm for Uncoiler & Recoiler coil car | 2 | Set |
| 6 | CRS | Boom barrier 1000 mm for Slitter arbour area | 1 | Set |
| 7 | CRS | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 8 | CRS | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| **Hot Rolled Slitter Line** | | | | |
| 1 | HRS | Safety Fence | 90 | Mtr |
| 2 | HRS | 2" MS Pipe barricaded structure | 42 | Mtr |
| 3 | HRS | Safety hinge door | 6 | Set |
| 4 | HRS | Safety Magnetic switch | 6 | Set |
| 5 | HRS | Boom barrier 3500 mm for Uncoiler & Recoiler coil car | 2 | Set |
| 6 | HRS | Boom barrier 1800 mm for Slitter arbour area | 1 | Set |
| 7 | HRS | Safety control panel with power supply & safety relay/PLC | 1 | Set |

| | | | | |
|---|---|---|---|---|
| 8 | HRS | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| **CR Wider Cut to length Line** | | | | |
| 1 | CRWCTL | Safety Fence | 100 | Mtr |
| 2 | CRWCTL | 2" MS Pipe barricaded structure | 17 | Mtr |
| 3 | CRWCTL | Safety hinge door | 5 | Set |
| 4 | CRWCTL | Safety Magnetic switch | 5 | Set |
| 5 | CRWCTL | Boom barrier 3000 mm for Uncoiler coil car | 1 | Set |
| 6 | CRWCTL | Boom barrier 3000 mm for Piler-1,2 | 4 | Set |
| 7 | CRWCTL | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 8 | CRWCTL | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| **CR Narrow Cut to length Line** | | | | |
| 1 | CRNCTL | Safety Fence | 83 | Mtr |
| 2 | CRNCTL | 2" MS Pipe barricaded structure | 15 | Mtr |
| 3 | CRNCTL | Safety hinge door | 5 | Set |
| 4 | CRNCTL | Safety Magnetic switch | 5 | Set |
| 5 | CRNCTL | Boom barrier 3000 mm for Uncoiler coil car | 1 | Set |
| 6 | CRNCTL | Boom barrier 2500 mm for Piler-1,2 | 4 | Set |
| 7 | CRNCTL | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 8 | CRNCTL | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| **HR Narrow Cut to length Line** | | | | |
| 1 | HRNCTL | Safety Fence | 90 | Mtr |
| 2 | HRNCTL | 2" MS Pipe barricaded structure | 17 | Mtr |
| 3 | HRNCTL | Safety hinge door | 5 | Set |
| 4 | HRNCTL | Safety Magnetic switch | 5 | Set |
| 5 | HRNCTL | Boom barrier 3000 mm for Uncoiler coil car | 1 | Set |
| 6 | HRNCTL | Boom barrier 2500 mm for Piler-1,2 | 4 | Set |
| 7 | HRNCTL | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 8 | HRNCTL | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| **Transfer trolley-1 & 2** | | | | |
| 1 | Transfer Trolley - 1 | 3" Heavy duty MS round pipe barricaded Structure with 1 meter in Height | 128 | Mtr |
| 2 | | Boom barrier 1000 mm for allowing crossing | 2 | Nos |
| 3 | | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 4 | | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| 1 | Transfer Trolley - 2 | 3" Heavy duty MS round pipe barricaded Structure with 1 meter in Height | 60 | Mtr |
| 2 | | Boom barrier 1000 mm for allowing crossing | 2 | Nos |
| 3 | | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 4 | | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| **Solar Inverter Panel Area** | | | | |
| 1 | Solar Inverter | Safety fencing with 1.5 Meter height with 2 Nos Hinge doors with lock & key mechanism | 40 | Mtr |

## BOQ FORMAT FOR DESIGN, ENGINEERING, SUPPLY, ERECTION, INSTALLATION, COMMISSIOING OF STEEL MESH SAFETY FENCING FOR PROCESS LINES OF PUNE SC

| SR. NO. | LINE | PRODUCT | QTY | UNIT |
|---|---|---|---|---|
| **Cold Rolled Slitter Line** | | | | |
| 1 | CRS | Steel Mesh Safety Fence | 50 | Mtr |
| 2 | CRS | 2" MS Pipe barricaded structure | 25 | Mtr |
| 3 | CRS | Safety hinge door | 1 | Set |
| 4 | CRS | Safety Sliding door | 2 | Set |
| 5 | CRS | Safety Magnetic switch | 3 | Set |
| 6 | CRS | Boom barrier 4000 mm for Uncoiler & Recoiler coil car | 2 | Set |
| 7 | CRS | Boom barrier 1000 mm for Slitter arbour area | 1 | Set |
| 8 | CRS | Boom barrier 1000 mm for Scrap Trolley | 1 | |
| 9 | CRS | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 10 | CRS | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| **Hot Rolled Slitter Line** | | | | |
| 1 | HRS | Safety Fence | 90 | Mtr |
| 2 | HRS | 2" MS Pipe barricaded structure | 20 | Mtr |
| 3 | HRS | Safety hinge door | 2 | Set |
| 4 | HRS | Safety Sliding door | 1 | Set |
| 5 | HRS | Safety Magnetic switch | 3 | Set |
| 6 | HRS | Boom barrier 4000 mm for Uncoiler & Recoiler coil car | 2 | Set |
| 7 | HRS | Boom barrier 1500 mm for Slitter arbour area | 1 | Set |
| 8 | HRS | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 8 | HRS | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| **CR Wider Cut to length Line** | | | | |
| 1 | CRWCTL | Safety Fence | 105 | Mtr |
| 2 | CRWCTL | 2" MS Pipe barricaded structure | 30 | Mtr |
| 3 | CRWCTL | Safety hinge door | 1 | Set |
| 4 | CRWCTL | Safety Sliding door | 2 | Set |
| 5 | CRWCTL | Safety Magnetic switch | 5 | Set |
| 6 | CRWCTL | Boom barrier 4000 mm for Uncoiler coil car | 1 | Set |
| 7 | CRWCTL | Boom barrier 3000 mm for Piler-1,2 | 4 | Set |
| 8 | CRWCTL | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 9 | CRWCTL | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| **CR Trapezoidal Cut to length Line** | | | | |
| 1 | CRTRAPCTL | Safety Fence | 95.9 | Mtr |
| 2 | CRTRAPCTL | 2" MS Pipe barricaded structure | 46.1 | Mtr |
| 3 | CRTRAPCTL | Safety hinge door | 2 | Set |
| 4 | CRTRAPCTL | Safety Sliding door | 1 | Set |

| | | | | |
|---|---|---|---|---|
| 5 | CRTRAPCTL | Safety Magnetic switch | 3 | Set |
| 6 | CRTRAPCTL | Boom barrier 3000 mm for Uncoiler coil car | 1 | Set |
| 7 | CRTRAPCTL | Boom barrier 3000 mm for Piler-1,2 | 4 | Set |
| 8 | CRTRAPCTL | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 9 | CRTRAPCTL | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| **CR Narrow Cut to length Line** | | | | |
| 1 | CRNCTL | Safety Fence | 50 | Mtr |
| 2 | CRNCTL | 2" MS Pipe barricaded structure | 21 | Mtr |
| 3 | CRNCTL | Safety hinge door | 1 | Set |
| 4 | CRNCTL | Safety Slinding door | 1 | Set |
| 5 | CRNCTL | Safety Magnetic switch | 2 | Set |
| 6 | CRNCTL | Boom barrier 3000 mm for Uncoiler coil car | 1 | Set |
| 7 | CRNCTL | Boom barrier 4000 mm for Piler-1,2 | 2 | Set |
| 8 | CRNCTL | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 9 | CRNCTL | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| **CR Mini Cut to length Line** | | | | |
| 1 | CR MINI CTL | Safety Fence | 30 | Mtr |
| 3 | CR MINI CTL | Safety hinge door | 2 | Set |
| 4 | CR MINI CTL | Safety Magnetic switch | 2 | Set |
| 5 | CR MINI CTL | Boom barrier 2000 mm for Uncoiler coil car | 1 | Set |
| 8 | CR MINI CTL | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 9 | CR MINI CTL | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| **HR Narrow Cut to length Line** | | | | |
| 1 | HRNCTL | Safety Fence | 80 | Mtr |
| 2 | HRNCTL | 2" MS Pipe barricaded structure | 24 | Mtr |
| 3 | HRNCTL | Safety hinge door | 4 | Set |
| 4 | HRNCTL | Safety Magnetic switch | 4 | Set |
| 5 | HRNCTL | Boom barrier 4000 mm for Uncoiler coil car | 1 | Set |
| 6 | HRNCTL | Boom barrier 2500 mm for Piler-1,2 | 4 | Set |
| 7 | HRNCTL | Boom barrier 4000 mm for Scrap Car Trolly | 1 | Set |
| 8 | HRNCTL | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 9 | HRNCTL | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| **HR Wider Cut to length Line** | | | | |
| 1 | HRWCTL | Safety Fence | 78 | Mtr |
| 2 | HRWCTL | 2" MS Pipe barricaded structure | 43 | Mtr |
| 3 | HRWCTL | Safety hinge door | 3 | Set |
| 4 | HRWCTL | Safety Magnetic switch | 3 | Set |
| 5 | HRWCTL | Boom barrier 4000 mm for Uncoiler coil car | 1 | Set |
| 6 | HRWCTL | Boom barrier 3000 mm for Leveller | 1 | Set |

| | | | | |
|---|---|---|---|---|
| 7 | HRWCTL | Boom barrier 2000 mm for Leveller | 1 | Set |
| 9 | HRWCTL | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 10 | HRWCTL | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| **Coil Transfer trolley** | | | | |
| | Coil Transfer Trolley | 3" Heavy duty MS round pipe barricaded Structure with 1 meter in Height | 74 | Mtr |
| | Coil Transfer Trolley | Boom barrier 2000 mm for allowing crossing | 2 | Set |
| | Coil Transfer Trolley | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| | Coil Transfer Trolley | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | Set |
| | | | | |
| 1 | Solar Inverter room | Safety fencing with 1.5 Meter height with 2 Nos Hinge doors with lock & key mechanism | 40 | Mtr |

## BOQ FORMAT FOR DESIGN, ENGINEERING, SUPPLY, ERECTION, INSTALLATION, COMMISSIOING OF STEEL MESH SAFETY FENCING FOR PROCESS LINES OF CHENNAI SC

| SR. NO. | LINE | PRODUCT | QTY | UNIT |
|---|---|---|---|---|
| **Cold Rolled Slitter Line** | | | | |
| 1 | CRS | Steel Mesh Safety Fence | 76 | Mtr |
| 2 | CRS | 2" MS Pipe barricaded structure | 32 | Mtr |
| 3 | CRS | Safety hinge door | 6 | Set |
| 4 | CRS | Safety Magnetic switch | 6 | Set |
| 5 | CRS | Boom barrier 3000 mm for Uncoiler & Recoiler coil car | 2 | Set |
| 6 | CRS | Boom barrier 1000 mm for Slitter arbour area | 1 | Set |
| 7 | CRS | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 8 | CRS | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| **Hot Rolled Slitter Line** | | | | |
| 1 | HRS | Safety Fence | 90 | Mtr |
| 2 | HRS | 2" MS Pipe barricaded structure | 42 | Mtr |
| 3 | HRS | Safety hinge door | 6 | Set |
| 4 | HRS | Safety Magnetic switch | 6 | Set |
| 5 | HRS | Boom barrier 3500 mm for Uncoiler & Recoiler coil car | 2 | Set |
| 6 | HRS | Boom barrier 1800 mm for Slitter arbour area | 1 | Set |
| 7 | HRS | Safety control panel with power supply & safety relay/PLC | 1 | Set |

| | | | | |
|---|---|---|---|---|
| 8 | HRS | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| **CR Wider Cut to Length Line** | | | | |
| 1 | CRWCTL | Safety Fence | 100 | Mtr |
| 2 | CRWCTL | 2" MS Pipe barricaded structure | 17 | Mtr |
| 3 | CRWCTL | Safety hinge door | 5 | Set |
| 4 | CRWCTL | Safety Magnetic switch | 5 | Set |
| 5 | CRWCTL | Boom barrier 3000 mm for Uncoiler coil car | 1 | Set |
| 6 | CRWCTL | Boom barrier 3000 mm for Piler-1,2 | 4 | Set |
| 7 | CRWCTL | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 8 | CRWCTL | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| **CR Narrow Cut to Length Line** | | | | |
| 1 | CRNCTL | Safety Fence | 83 | Mtr |
| 2 | CRNCTL | 2" MS Pipe barricaded structure | 15 | Mtr |
| 3 | CRNCTL | Safety hinge door | 5 | Set |
| 4 | CRNCTL | Safety Magnetic switch | 5 | Set |
| 5 | CRNCTL | Boom barrier 3000 mm for Uncoiler coil car | 1 | Set |
| 6 | CRNCTL | Boom barrier 2500 mm for Piler-1,2 | 4 | Set |
| 7 | CRNCTL | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 8 | CRNCTL | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| **HR Narrow Cut to Length Line** | | | | |
| 1 | HRNCTL | Safety Fence | 90 | Mtr |
| 2 | HRNCTL | 2" MS Pipe barricaded structure | 17 | Mtr |
| 3 | HRNCTL | Safety hinge door | 5 | Set |
| 4 | HRNCTL | Safety Magnetic switch | 5 | Set |
| 5 | HRNCTL | Boom barrier 3000 mm for Uncoiler coil car | 1 | Set |
| 6 | HRNCTL | Boom barrier 2500 mm for Piler-1,2 | 4 | Set |
| 7 | HRNCTL | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 8 | HRNCTL | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| | | | | |
| 1 | HRWCTL | Safety Fence | 80 | Mtr |
| 2 | HRWCTL | 2" MS Pipe barricaded structure | 18 | Mtr |
| 3 | HRWCTL | Sliding Doors of 1000 mm Width | 5 | Nos |
| 4 | HRWCTL | Hinged Doors of 800 mm width | 2 | Nos |
| 5 | HRWCTL | Boom Barrier of 3500 mm width | 1 | No |
| 6 | HRWCTL | Light Curtain (Leveller movement, Width 2000mm | 1 | No |
| **Coil Transfer trolley** | | | | |
| 1 | Coil Transfer Trolley | 3" Heavy duty MS round pipe barricaded Structure with 1 meter in Height | 230 | Mtr |

| | | | | |
|---|---|---|---|---|
| 2 | Coil Transfer Trolley | Boom barrier 1000 mm for allowing crossing | 2 | Set |
| 3 | Coil Transfer Trolley | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 4 | Coil Transfer Trolley | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | Set |
| **Solar Inverter Panel Area** | | | | |
| 1 | Solar Inverter Panels | Safety fencing with 1.5 Meter height with 2 Nos Hinge doors with lock & key mechanism | 40 | Mtr |

## BOQ FORMAT FOR DESIGN, ENGINEERING, SUPPLY, ERECTION, INSTALLATION, COMMISSIOING OF STEEL MESH SAFETY FENCING FOR PROCESS LINES OF INDORE SC

| SR. NO. | LINE | PRODUCT | QTY | UNIT |
|---|---|---|---|---|
| **Hot Rolled Slitter Line** | | | | |
| 1 | HRS | Safety Fence | 90 | Mtr |
| 2 | HRS | 2" MS Pipe barricaded structure | 15 | Mtr |
| 3 | HRS | Safety hinge door | 7 | Set |
| 4 | HRS | Safety Magnetic switch | 7 | Set |
| 5 | HRS | Boom barrier 3500 mm for Uncoiler & Recoiler coil car | 2 | Set |
| 6 | HRS | Boom barrier 1800 mm for Slitter arbour area | 1 | Set |
| 7 | HRS | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 8 | HRS | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| **HR Wider Cut to length line** | | | | |
| 1 | HRWCTL | Safety Fence | 105 | Mtr |
| 3 | HRWCTL | Safety hinge door | 7 | Set |
| 4 | HRWCTL | Safety Magnetic switch | 7 | Set |
| 5 | HRWCTL | Boom barrier 3500 mm for Uncoiler coil car | 1 | Set |
| 6 | HRWCTL | Boom barrier 1800 mm for Leveller movement | 1 | Set |
| 7 | HRWCTL | Light Curtain 8000 mm range with protection guard | 2 | Set |
| 8 | HRWCTL | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 9 | HRWCTL | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | LOT |
| **Coil Transfer Trolley** | | | | |
| 1 | Coil Transfer Trolley | 3" Heavy duty MS round pipe barricaded Structure with 1 meter in Height | 40 | Mtr |
| 2 | Coil Transfer Trolley | Boom barrier 1000 mm for allowing crossing | 2 | Set |

| | | | | |
|---|---|---|---|---|
| 3 | Coil Transfer Trolley | Safety control panel with power supply & safety relay/PLC | 1 | Set |
| 4 | Coil Transfer Trolley | Cabling, Thimbles, conduit, glands, indicators, hooters & required accessories for dressing & fixing | 1 | Set |

**AMNS SAFETY RULE AND REGULATION AS PER COMPANY POLICY**

1) All necessary safety PPE's i.e. safety shoes, leather gloves, goggles, masks etc. in contractor scope
2) All work to be carried out as per safety norms which shall be guided to AM/NS team during work & no tolerance in safety will be accepted and penalty to be imposed if any violation.
3) Contractor shall not proceed with next activity until previous activity will be checked and approved by Representative.
4) All required tools & tackles will provide by contractor
5) Contractor shall deploy and provide one full time engineer to supervise the project and has experienced of carrying out such type of work.
6) Contractor will provide all required consumable items for the job
7) Contractor shall be responsible for procuring, supplying, transporting, and providing all labor, materials, tools and equipment etc., required for completion of the work.
8) All materials shall be stored in a proper manner protected from natural elements so as to avoid contamination and deterioration.
9) Workers working on the site shall be skilled in their job and have related job experience.
10) The contractor shall guarantee that all work performed will be free from all defects in workmanship and materials and that all installation will provide the capacities and characteristics specified. The contract further guarantees that if, during a period of one year from the date of the certificate of completion and acceptance of the work, any such defects will be repaired by the contractor at his own cost.

**AMNS Scope:**

1) Electricity supply
2) On site help i.e. Workshop Facilities etc. in AM/NS scope